

ECO387C - Fall 2024

Problem Set 9 Solutions

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Two-period endowment economy with government

Part 1: Representative agent There are two periods: period 0 and period 1, and one good. There is only one type of agent, a household. Assume that the total mass of agents is $N = 1$. In period 0, each agent seeks to maximize the expected lifetime utility.

$$\max \sum_{t=0}^1 \beta^t u(c_t)$$

subject to the budget constraints

$$\begin{aligned} c_0 + b_1^h &= y_0 - \tau_0 \\ c_1 &= y_1 + (1+r)b_1^h - \tau_1 \end{aligned}$$

where b^h is the risk-free bond owned by households. τ is the tax imposed by the government. Tax revenues collected by the government are used to pay for government consumption and to service the government debt. The government complies with the following budget constraints:

$$\begin{aligned} g_0 + b_1^g &= \tau_0 \\ g_1 &= \tau_1 + (1+r)b_1^g \end{aligned}$$

where g is the government expenditure. Note that g and τ are considered exogenous.

1. Set up the Lagrangian and derive the Euler equation of the household.

The Euler equation is

$$u'(c_0) = \beta u'(c_1)(1+r)$$

2. Derive the intertemporal budget constraint for the household

$$c_0 + \frac{1}{1+r}c_1 = y_0 + \frac{1}{1+r}y_1 - \tau_0 - \frac{1}{1+r}\tau_1$$

3. Derive the intertemporal budget constraint for the government

$$g_0 + \frac{1}{1+r}g_1 = \tau_0 + \frac{1}{1+r}\tau_1$$

4. Suppose $u(c) = \log c$, find the optimal consumption policy functions of households.

From the Euler equation,

$$\beta(1+r)c_0 = c_1$$

Using the intertemporal budget constraint

$$\begin{aligned} c_0(1+\beta) &= y_0 + \frac{1}{1+r}y_1 - \tau_0 - \frac{1}{1+r}\tau_1 \\ c_0^* &= \frac{(y_0 - \tau_0) + \frac{1}{1+r}(y_1 - \tau_1)}{1+\beta} \\ c_1^* &= \frac{\beta(1+r)((y_0 - \tau_0) + \frac{1}{1+r}(y_1 - \tau_1))}{1+\beta} \end{aligned}$$

5. Consider the definition of equilibrium below:

Equilibrium: Given $\{g_t\}_{t=0}^1$ and $\{\tau_t\}_{t=0}^1$ that respect the intertemporal budget constraint of the government, an equilibrium in this economy consists a sequence of allocations $\{c_0, c_1, b_1^h, b_1^g\}$ and price r such that

- (a) Taking price as given, the allocation solves the household's maximization problem.
- (b) Markets clear
 - Goods market

$$c_t + g_t = y_t \text{ for } t = 0, 1$$

- Asset market

$$b_1^h + b_1^g = 0$$

Given the definition of equilibrium, suppose that the endowment follows $(y_0, y_1) = (5, 5)$. The government sets up a consumption plan $(g_0, g_1) = (1, 0)$. To finance this, the government is considering the two tax schemes below. Assume $\beta = 1$ and find each scheme's equilibrium allocation and interest rates. Does the equilibrium consumption allocation depend on the tax scheme? In other words, does the Ricardian equivalence hold?

- (a) $(\tau_0, \tau_1) = (1, 0)$:

The optimal consumption in each period is

$$\begin{aligned} c_0^* &= \frac{(y_0 - \tau_0) + \frac{1}{1+r}(y_1 - \tau_1)}{1+\beta} = \frac{(5-1) + \frac{1}{1+r}(5-0)}{1+1} = \frac{4+5(1+r)^{-1}}{2} \\ c_1^* &= \frac{\beta(1+r)((y_0 - \tau_0) + \frac{1}{1+r}(y_1 - \tau_1))}{1+\beta} = \frac{(1+r)(4+5(1+r)^{-1})}{2} \end{aligned}$$

Using the goods market clearing condition, we can obtain the equilibrium interest rate $(1+r^*) = \frac{5}{4}$. We can use this to back out optimal consumption levels in each period. Furthermore, we can compute the optimal debt possessed by the households and the government using their budget constraint. The equilibrium allocation and prices are

$$(c_0^*, c_1^*, b_1^{h*}, b_1^{g*}, 1+r^*) = (4, 5, 0, 0, \frac{5}{4})$$

(b) $(\tau_0, \tau_1) = (0, 1+r)$:

The optimal consumption in each period is

$$c_0^* = \frac{(y_0 - \tau_0) + \frac{1}{1+r}(y_1 - \tau_1)}{1 + \beta} = \frac{5 + \frac{1}{1+r}(5 - (1+r))}{1 + 1} = \frac{4 + 5(1+r)^{-1}}{2}$$

$$c_1^* = \frac{\beta(1+r)((y_0 - \tau_0) + \frac{1}{1+r}(y_1 - \tau_1))}{1 + \beta} = \frac{(1+r)(4 + 5(1+r)^{-1})}{2}$$

Using the goods market clearing condition, we can obtain the equilibrium interest rate $(1+r^*) = \frac{5}{4}$. We can use this to back out optimal consumption levels in each period. Furthermore, we can compute the optimal debt possessed by the households and the government using their budget constraint. The equilibrium allocation and prices are

$$(c_0^*, c_1^*, b_1^{h*}, b_1^{g*}, 1+r^*) = (4, 5, 1, -1, \frac{5}{4})$$

6. Provide an interpretation of the result: Allocation of consumption is indifferent to how the government expenditure is financed. Ricardian equivalence holds as long as the tax is imposed such that the present discounted value matches that of the government expenditure.

Part 2: Hand-to-mouth agent Assume that the total mass of agents is still $N = 1$. However, there are now two types of agents. The first type has unconstrained access to borrowing as in the previous part. Their budget constraints are

$$c_{u,0} + b_1^h = y_{u,0} - \tau_0$$

$$c_{u,1} = y_{u,1} + (1+r)b_1^h - \tau_1$$

The second type of household is constrained in the sense that it does not have access to borrowing. Their budget constraints are

$$c_{c,0} = y_{c,0} - \tau_0$$

$$c_{c,1} = y_{c,1} - \tau_1$$

Both have the same objective function as in the previous part. The fraction of each type is $\frac{1}{2}$.

1. Suppose $u(c) = \log(c)$. What is the optimal consumption policy function for each type of agent?

- Unconstrained:

$$c_{u,0}^* = \frac{(y_0 - \tau_0) + \frac{1}{1+r}(y_1 - \tau_1)}{1 + \beta}$$

$$c_{u,1}^* = \frac{\beta(1+r)((y_0 - \tau_0) + \frac{1}{1+r}(y_1 - \tau_1))}{1 + \beta}$$

- Constrained:

$$c_{c,0}^* = y_0 - \tau_0$$

$$c_{c,1}^* = y_1 - \tau_1$$

2. Consider the following endowment process and government expenditure plan: $(y_{u,0}, y_{u,1}) = (y_{c,0}, y_{c,1}) = (5, 5)$ and $(g_0, g_1) = (1, 0)$. Assume $\beta = 1$ and find the equilibrium allocation and interest rates in the following two tax schemes. Does the Ricardian equivalence hold?

(a) $(\tau_0, \tau_1) = (1, 0)$

The optimal consumption for the unconstrained type is

$$c_{u,0}^* = \frac{(y_0 - \tau_0) + \frac{1}{1+r}(y_1 - \tau_1)}{1 + \beta} = \frac{(5 - 1) + \frac{1}{1+r}(5 - 0)}{1 + 1} = \frac{4 + 5(1+r)^{-1}}{2}$$

$$c_{u,1}^* = \frac{\beta(1+r)((y_0 - \tau_0) + \frac{1}{1+r}(y_1 - \tau_1))}{1 + \beta} = \frac{(1+r)(4 + 5(1+r)^{-1})}{2}$$

The optimal consumption for the constrained type is

$$c_{c,0}^* = y_0 - \tau_0 = 5 - 1 = 4$$

$$c_{c,1}^* = y_1 - \tau_1 = 5 - 0 = 5$$

Using the goods market clearing condition:

$$\frac{1}{2}c_{u,0}^* + \frac{1}{2}c_{c,0}^* + g_0 = y_0 \Leftrightarrow \frac{1}{2} \frac{4 + 5(1+r)^{-1}}{2} + \frac{1}{2}4 + 1 = 5$$

we can obtain the equilibrium interest rate $(1+r^*) = \frac{5}{4}$. We can use this to back out optimal consumption levels in each period. Furthermore, we can compute the optimal debt possessed by the households and the government using their budget constraint. The equilibrium allocation and prices are

$$(c_{u,0}^*, c_{u,1}^*, c_{c,0}^*, c_{c,1}^*, b_1^{h*}, b_1^{g*}, 1+r^*) = (4, 5, 4, 5, 0, 0, \frac{5}{4})$$

(b) $(\tau_0, \tau_1) = (0, 1+r)$

The optimal consumption for the unconstrained type is

$$c_{u,0}^* = \frac{(y_0 - \tau_0) + \frac{1}{1+r}(y_1 - \tau_1)}{1 + \beta} = \frac{5 + \frac{1}{1+r}(5 - (1+r))}{1 + 1} = 2 + \frac{5}{2(1+r)}$$

$$c_{u,1}^* = \frac{\beta(1+r)((y_0 - \tau_0) + \frac{1}{1+r}(y_1 - \tau_1))}{1 + \beta} = 2(1+r) + \frac{5}{2}$$

The optimal consumption for the constrained type is

$$c_{c,0}^* = y_0 - \tau_0 = 5 - 0 = 5$$

$$c_{c,1}^* = y_1 - \tau_1 = 5 - (1+r) = 4 - r$$

Then, using the market clearing condition:

$$\frac{1}{2}c_{u,0}^* + \frac{1}{2}c_{c,0}^* + g_0 = y_0 \Leftrightarrow \frac{1}{2} \left(2 + \frac{5}{2(1+r)} \right) + \frac{1}{2}5 + 1 = 5$$

This implies $1+r^* = \frac{5}{2}$. The equilibrium is

$$(c_{u,0}^*, c_{u,1}^*, c_{c,0}^*, c_{c,1}^*, b_1^{h*}, b_1^{g*}, 1+r^*) = (3, \frac{15}{2}, 5, \frac{5}{2}, 2, -1, \frac{5}{2})$$

3. Provide an interpretation of the result: Consumption allocation across different types of households depends on how government spending is financed. Thus, Ricardian equivalence fails.